Design of an AI-Based Litter Inspection and Pickup Robot for Urban Expressway Median Strips: Project Report

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Shanxi Provincial Undergraduate Innovation Training Program, Project No. S202514527040 (May 2025 – May 2026)

***Abstract*—Litter thrown from vehicle windows accumulates in the central median strips of urban expressways. It degrades the cityscape and creates fire and visual hazards, and removing it manually exposes sanitation workers to heat and fast traffic and often requires lane closures. This report presents the design of an unmanned litter inspection and pickup robot for expressway medians. We first surveyed the full 2.1 km of Yantai Street in Yangquan, China, characterizing the median geometry, traffic patterns, and litter composition: plastics account for 42% of items, paper for 28%, and cigarette butts for 15%, at an average density of 3.9 items/m^2 in the median. Based on these findings we designed (i) a rear-wheel-drive, front-wheel-steer locomotion mechanism with rack-and-pinion steering, including motor sizing and shaft strength verification; (ii) a planar articulated pickup arm with a 1.2 m reach and a dual gripper/suction end effector; and (iii) an edge–cloud control platform that combines YOLO-based litter detection in the cloud with ESP32-S3 real-time control on the robot, GIS visualization, and remote task management. A YOLO11n detector trained for 200 epochs on a 17-class litter dataset reaches 0.864 mAP@0.5 and 0.583 mAP@0.5:0.95 on the validation set, with a peak F1 of 0.84. Most of its errors are missed detections rather than confusions between classes, and leaves and glass bottles remain the weakest classes. The control software (V1.0) has been completed, while a physical prototype and field trials remain future work.**

***Index Terms*—Litter pickup robot, expressway median, object detection, YOLO, edge–cloud control, mobile manipulator, mechanism design.**

I. Introduction

Expressways are key arteries of urban transport, and their condition reflects directly on city management. Litter thrown from vehicle windows, such as bottles, food wrappers, tissues, and cigarette butts, collects in the central median strip. It pollutes the environment, produces odors, poses fire risks, and distracts drivers.

Today this litter is collected by hand. Manual cleaning is slow and costly, and sanitation workers must operate in summer heat next to fast-moving traffic, which exposes them to serious danger. To protect workers, lanes are often closed in advance, which adds to traffic pressure. An unmanned robot that can patrol the median and pick up litter autonomously and in all conditions would address all three problems.

This project, funded under the Shanxi Provincial Undergraduate Innovation Training Program, set out to develop such a robot. The robot consists of a mechanical system and a control system. The mechanical system includes an unmanned mobile platform, a pickup arm, a litter storage module with compaction, and a ground-cleaning mechanism. The control system covers power, obstacle avoidance and path following, litter recognition, arm control, operation warnings, bin-capacity alerts, positioning, and remote operation. Three technical problems were identified as key:

(A) an obstacle-avoiding, path-following locomotion system;
(B) a deep-learning litter recognition system;
(C) the structure of the pickup arm.

This report describes the work completed during the project period. After a review of related work and our research methodology (Section III), it presents two field surveys (Section IV), the overall system concept (Section V), the locomotion mechanism (Section VI), the pickup mechanism (Section VII), the control platform (Section VIII), the perception and control algorithms (Section IX), the training and evaluation of the litter detector (Section X), and safety functions (Section XI). We then summarize design targets and the test plan (Section XII), project outcomes (Section XIII), and limitations and future work.

Scope. The project produced survey reports, mechanism designs, and working control software. No physical prototype was built during the project period, so the performance figures in this report are design targets or calculated values unless stated otherwise.

II. Related Work

A. Litter Detection

Vision-based litter detection has developed alongside general object detection. Early work classified single, centered waste items in controlled images, as in the TrashNet dataset [2]. Litter in the wild is harder: items are small, deformed, partly occluded, and set against cluttered natural backgrounds. The TACO dataset [3] addresses this with annotated photographs of litter in real environments and highlights how much detection performance drops for small and occluded objects. One-stage detectors of the YOLO family [1,5–7] are widely used for such tasks because they combine competitive accuracy with real-time speed. Feature pyramids [4] help detect objects across a wide range of scales.

Our survey shows that expressway medians present exactly these difficulties. Cigarette butts about 1 cm long make up 15% of items, and many plastic and paper items hang in shrubs or hide under leaves. This motivated a dataset built from the target scene rather than reliance on a generic dataset alone.

B. Mobile Manipulation for Cleaning

A litter-picking robot is a mobile manipulator. A wheeled base carries an arm, and perception must guide both. Two problems are central. The first is converting image detections into arm targets, which requires camera calibration [9] and a hand–eye transform. The second is solving the arm's inverse kinematics within joint limits [10, 11]. Low-cost microcontrollers such as the ESP32 family make it practical to run real-time motor control on the robot while offloading compute-heavy detection to a server. This edge–cloud split, adopted in our design, trades network dependence for lower on-board cost and power.

C. Positioning of This Work

Compared with general-purpose cleaning robots, this project focuses on one specific, hazardous setting: the narrow, vegetated median strip between fast traffic lanes. The contribution of the project is therefore scenario-driven. It quantifies the environment and litter on a real expressway, derives requirements from them, and designs the mechanism, perception, and management platform to meet those requirements.

III. Research Methodology

The project followed a combination of theoretical analysis, computer simulation, and experimental validation. Fig. 1 shows the technical route, which has eight steps:

1. *Field survey.* Investigate urban roads to understand the requirements of median cleaning, analyze the robot's functional needs, and draft a design concept.
2. *CAD modeling.* Build a virtual prototype of the mechanical structure in SolidWorks.
3. *Kinematic design.* Simulate arm motion in MATLAB and parameterize the mechanisms.
4. *Dynamic design.* Analyze the dynamics of the selected structures.
5. *Structural analysis.* Run finite-element analysis in ANSYS to optimize the design.
6. *Recognition and control.* Design the path-following, obstacle-avoidance, and litter-recognition systems: a YOLO model identifies targets and road conditions and locates litter, and PID control drives the vehicle and arm.
7. *Prototyping.* Develop manufacturing processes and build a prototype, then run mechanical and motion tests and optimize parameters and structure.
8. *Field trials and deployment.* Test on site, refine the structure, and move to small-batch production and deployment.

Steps 1–2 and 6 were completed during the project period. Steps 3–5 were carried out as analytical calculations in the design documents (Sections VI and VII); simulation and finite-element analysis remain to be done. Steps 7–8 are future work.

IV. Field Survey

To ground the design in real conditions, the team surveyed the full length of Yantai Street in the Suburban District of Yangquan, Shanxi, on two occasions in June 2025. The first survey (June 1) recorded road geometry, the median layout, vegetation, obstacles, and traffic at midday (11:30–13:00) and in the evening peak (17:30–19:00). The second (June 3) catalogued litter types and their spatial distribution through a walking census of every green belt along the road.

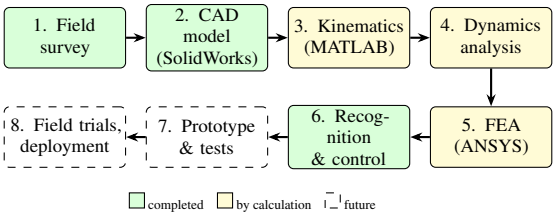


Figure 1. Technical route of the project and the status of each step.

Table I
Median Strip Characteristics of Yantai Street

Item	Value
Road length / lanes	2.1 km, two-way four lanes
Design speed	60 km/h
Median width	1.8–2.2 m (mean 2.0 m)
Height (curb to canopy)	0.2–1.2 m
Openings	1.5 m wide, every 50 m
Vegetation cover	90% (trees, shrubs, lawn)
Drain covers / manholes	every 20 m / every 30 m
Local unevenness	$\leq$0.3 m
Longitudinal slope	1.2–2.5%

A. Site Characteristics

Yantai Street runs east–west for about 2.1 km. It is a two-way, four-lane road with a design speed of 60 km/h. Straight sections make up 85% of its length, with two gentle curves (radius $\geq$500 m) and no steep grades. Longitudinal slopes are 1.2–2.5% and the cross slope is 2%. Table I summarizes the median. It is 1.8–2.2 m wide (2.0 m on average), bounded by 0.2 m concrete curbs, and has a 1.5 m opening every 50 m for U-turns and pedestrian crossings. Vegetation covers 90% of the median. Pagoda trees are planted at 4 m spacing, with shrubs such as purple-leaf plum and golden privet beneath them and a bluegrass lawn at ground level. Storm-drain covers (0.7 m diameter, every 20 m) and sewer manholes (0.8 m, every 30 m) lie in or beside the green belt, and the ground has local bumps and depressions of up to 0.3 m.

B. Traffic

Small passenger cars dominated traffic in both periods, at about 85% of vehicles at midday and 88% in the evening. Midday traffic was light and moved at 40–50 km/h, whereas the evening peak brought markedly higher vehicle, cyclist, and pedestrian volumes with occasional slow-moving traffic. Pedestrians crossing at median openings and at residential entrances on the south side were identified as the main dynamic hazards for a robot.

Table II breaks the observations down by vehicle class. Medium vehicles were mostly delivery vans serving nearby shops, and large vehicles were buses and a few through trucks. Cyclists kept to the side lanes at 15–25 km/h. In the evening, cyclist and pedestrian volumes rose more sharply than vehi-

Table II
TRAFFIC OBSERVATIONS BY PERIOD

	Midday (11:30–13:00)	Evening (17:30–19:00)
Overall volume	Low, free-flowing	Peak, occasional slow traffic
Small cars (share, speed)	85%, 40–50 km/h	88%, 35–45 km/h
Medium vehicles	10%, 35–45 km/h	9%, 30–40 km/h
Large vehicles	5%, 30–40 km/h	3%, 25–35 km/h
Cyclists	Few, 15–20 km/h	Many, 15–25 km/h
Pedestrians	Moderate to low	High; frequent crossings
Recommended operation	Full routine patrol	Emergency or zoned patrol

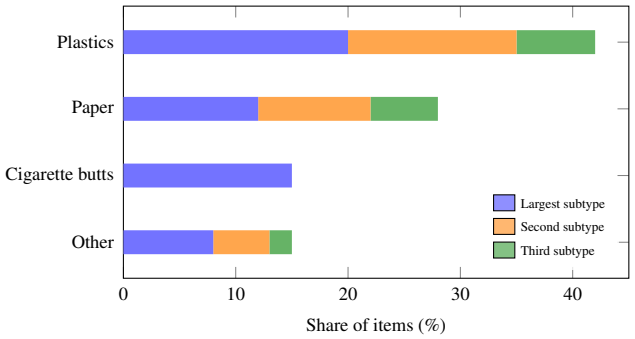


Figure 2. Litter composition on Yantai Street. Subtypes, in order: plastics (bags/wrappers, PET bottles, containers/straws/caps); paper (tissues, flyers/receipts, boxes/cups); other (fruit peel, leaves/weeds, pet waste).

cle volumes, as residents walked and commuted along the sidewalks and crossed at median openings much more often. Traffic was orderly in both periods, with no sudden surges, which suits a low-speed patrol robot. The survey therefore recommends routine patrols at midday, when disturbance is low, and only emergency patrols or patrols restricted to the quieter north side during the evening peak.

C. Litter Composition and Distribution

Fig. 2 shows the litter composition. Plastics are the largest category at 42% of items: soft bags and wrappers 20%, PET bottles 15%, and containers, straws, and caps 7%. Paper follows at 28%: tissues 12%, flyers and receipts 10%, and boxes and cups 6%. Cigarette butts account for 15%, and other litter for 15% (fruit peel 8%, leaves and weeds 5%, pet waste 2%). The average density in the central median is 3.9 items/m^2, made up of plastics 1.6, paper 1.1, butts 0.7, and other 0.5. The side green belts are dirtier: 6.5 items/m^2 on the separator next to the bike lane and 6.8 items/m^2 in tree pits, where butts alone reach 2.9 items/m^2.

Litter in the median follows a clear spatial pattern. Within 0–50 m of intersections and stop lines, litter is 3–5 times denser than elsewhere, dominated by butts, tissues, and small bags thrown by occupants of cars waiting at red lights. On straight sections (50–200 m), litter is scattered evenly, mostly bottles and wrappers, because throws from cars traveling at 60–80 km/h land in dispersed positions. Beyond 200 m, at curves and slopes, totals are lower but items tend to hide in shrubs or lawn. Vertically, most plastic and paper items hang in the shrub layer (0.5–1.2 m), while butts and paper scraps lie on the lawn (0–0.5 m). The team attributes about 80% of median litter to car-window littering, 15% to pedestrians, and 5% to wind and leaf cover.

Table III
LITTER DENSITY BY LOCATION (ITEMS/M^2)

Location	Total	Plastic	Paper	Butts	Other
Central median	3.9	1.6	1.1	0.7	0.5
Bike-lane separator	6.5	–	–	–	–
Tree pits	6.8	–	–	2.9	–

Table III summarizes litter density by location. The side belts collect mixed litter from pedestrians and vehicles; the separator next to the bike lane holds about 1.8 times as much litter as the central median, mainly food bags, fruit peel, and butts. Tree pits and hedge gaps trap fine litter such as butts and paper scraps. About 75% of cigarette butts were embedded in soil or leaf litter, and up to 120 butts were counted along the road, which makes them both the most numerous and the hardest target. Paper becomes fragile and decays when wet, and in damp conditions it blends with leaves and soil.

D. Implications for Design

Table IV translates the survey findings into design requirements. For the recognition model, the survey points to three hard cases that need deliberate attention in data collection and annotation: small targets (1 cm cigarette butts), deformed items hanging in shrubs, and items partly hidden by leaves or grass. It also shows that object sizes range from about 1 cm to 30 cm.

The survey shapes the dataset in four ways:

- *Category system.* Small butts, deformed hanging plastics, and occluded items receive dedicated attention, since they are the most likely to be missed or misclassified.
- *Scene coverage.* Samples should span intersections at peak hours, straight sections, and hidden spots at curves.
- *Annotation detail.* Labels should record size (from 1 cm butts to 30 cm bottles), state (intact, broken, deformed), and occlusion.
- *Sampling proportions.* The measured category shares and densities guide how many samples to collect per class.

For the locomotion and safety systems, the survey recommends a four-wheel ground footprint that copes with lawn, bare soil, and paving. It also recommends 360° LiDAR combined with an HD camera and ultrasonic sensors covering 0.1–10 m, able to recognize pedestrians, cyclists, vehicles, poles, and manhole covers and to predict the paths of moving obstacles. Further recommendations are high-precision maps with preset patrol routes and resume-after-interruption, a wear-resistant shell with soft edges, and lights and voice prompts asking pedestrians to give way.

V. SYSTEM OVERVIEW

Fig. 3 shows the 3D model of the robot. A four-wheeled chassis carries an articulated pickup arm at the front and storage bins on the deck. The arm ends in a gripper for rigid items and a suction cup for light, flexible items. Functionally, the robot follows a “move–stop–pick–move” cycle. It patrols the

Table IV
DESIGN REQUIREMENTS DERIVED FROM THE SURVEY

Finding	Requirement
Gentle grades, local bumps $\leq$0.3 m, dense manholes	Wheeled base with suspension; obstacle crossing $\geq$0.1 m
Narrow median, shrubs 0.8–1.2 m	Width $\leq$1.4 m, height $\leq$1.7 m; sensors mounted at 1.0–1.5 m
Pedestrians at openings, cyclists	Obstacle response $\leq$0.4–0.5 s; yield to pedestrians; speed $\leq$4 km/h
Adjacent traffic at 30–50 km/h	Keep $\geq$0.5 m from lane edge; lights, reflectors, audible warnings
Evening peak congestion	Full patrol at midday; zoned or emergency-only work at peak
Fixed poles and manholes	Pre-load fixed obstacles into the map for path planning
Midday glare, dusk low light	Sensors robust to strong and back light; -10 to 40 °C operation
Litter in shrub layer, median $\approx$2 m wide	Arm able to reach across the median and into shrubs

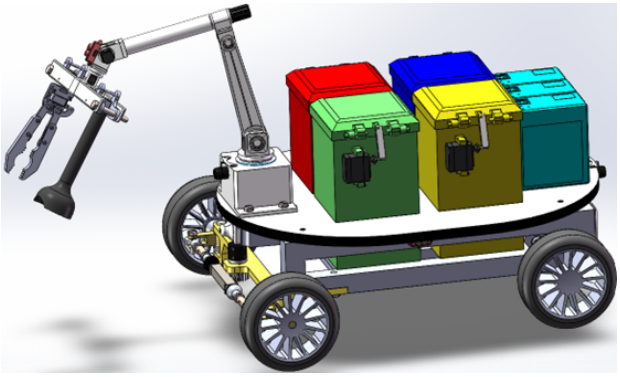


Figure 3. 3D model of the litter inspection and pickup robot: four-wheeled chassis, articulated pickup arm with gripper and suction end effectors, and on-board storage bins.

median along a planned path; when the camera detects litter within the arm's reach, it stops, picks up the item, deposits it in a bin, and resumes patrol. Obstacles detected along the way trigger steering maneuvers. All detections, positions, and robot states stream to a cloud platform for monitoring and task management.

The project produced two levels of design. Sections VI and VII describe a *full-scale field design* sized for the expressway median, with a total mass of about 400 kg and an arm reach of 1.2 m. Section VIII describes the control platform together with a *reduced-scale control prototype* specification built from low-cost embedded hardware. The prototype is intended to validate perception, control, and edge–cloud coordination before the full-scale machine is built.

VI. LOCOMOTION MECHANISM

A. Requirements

The locomotion mechanism must move the robot steadily along the median, steer around obstacles, and stop precisely for pickup. It was designed for a maximum speed of 1 m/s, front-wheel steering of $\pm 30^\circ$, and a total load of about 400 kg (robot body, arm, and bins) without structural deformation, with reliable braking, and with a structure that is simple to machine, assemble, and maintain.

Table V
COMPARISON OF LOCOMOTION SCHEMES

	Rear drive, front steer	4WD, 4-wheel steer
Motors	2 drive + 1 steering	4 drive + 4 steering
Structure	Simple, easy to build	Complex
Control	Simple	Difficult
Cost	Lower	High
Suitability	Meets patrol and avoidance needs	Redundant for median patrol
Decision	**Selected**	Rejected

B. Scheme Selection

Two schemes were compared (Table V). Four-wheel drive with four-wheel steering offers the most maneuverability but needs four drive and four steering motors, and median patrol does not require such complex maneuvers. Rear-wheel drive with front-wheel steering needs only two drive motors and one steering motor, keeps the control logic simple, and meets the requirements for straight travel and obstacle avoidance. We therefore selected rear-wheel drive with front-wheel steering.

C. Working Principle

When the robot starts, the rear drive motors move it forward at constant speed while a LiDAR scans the surroundings. If no litter is within the arm's reach, the robot keeps moving and searching. When litter is found within reach, it stops, the pickup mechanism collects the item and places it in the bin, and the robot then resumes travel.

For obstacle avoidance, the front LiDAR monitors the path ahead and side sensors monitor the flanks. If an obstacle is detected ahead while the side is clear, the steering motor turns the front wheels. The rear wheels keep driving during the turn to maintain a smooth, constant speed and to avoid jerks or tipping. After the maneuver, the front wheels return to straight ahead.

D. Drive Motor Sizing

Braking. With a disc brake, the braking force is

$$F_b = 2fN, \qquad N = pA, \tag{1}$$

where $f = 0.4$ is the friction coefficient, $p = 8$ MPa the wheel-cylinder pressure, and A the area of a 32 mm piston ($A \approx 804.25\,\text{mm}^2$). This gives $N \approx 6434$ N and $F_b \approx 5147$ N, a deceleration capacity of $F_b/m \approx 12.9\,\text{m/s}^2$ for $m = 400$ kg. The maximum speed was set to 1 m/s with safety on the median in mind.

Drive torque. Assuming the wheels roll without slipping (Fig. 4), force and moment balance about the rear axle give the drive torque

$$M_d = F_d r + M_f, \qquad M_f = \delta mg, \tag{2}$$

with wheel radius $r = 0.15$ m and rolling-resistance coefficient $\delta = 0.007$ m. The calculation yields $M_d = 27.44$ N·m.

Speed and power. At $v = 1$ m/s the wheel angular speed is $\omega = v/r \approx 6.67$ rad/s, i.e., $n \approx 95.5$ r/min. With a safety factor $K = 2$, the required power is

$$P = \frac{K M_d n}{9550} \approx 0.549\ \text{kW}. \tag{3}$$

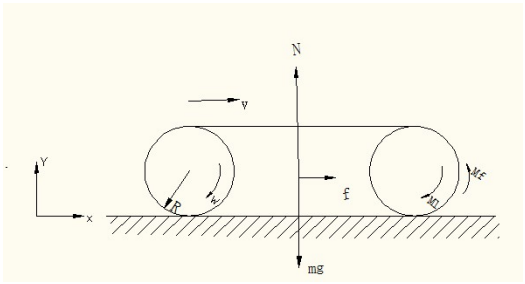


Figure 4. Free-body diagram of the chassis during travel: ground reactions N, rolling resistance f, weight mg, and drive moment on the wheels.

Table VI
STEERING MECHANISM PARAMETERS

Part	Parameter	Value
Rack	Length / diameter	730 mm / 25 mm
	Teeth / normal module	31 / 2.5 mm
Pinion	Teeth / normal module	18 / 2.5 mm
	Pressure angle	20°
Tie rod	Length / diameter	213 mm / 16 mm
	Thread length	30 mm
Joints	Outer / inner length	120 / 65.3 mm
	Ball stud length, thread	62 mm, M10×1
Mesh	Engagement length	100 mm

The overall efficiency is $\eta_\Sigma = \eta_m \eta_t \eta_w = 0.87 \times 0.80 \times 0.92 \approx 0.639$, combining motor, traction, and transmission efficiency. The motor output power must therefore be $P/\eta_\Sigma \approx 859$ W. We selected a WZDA-5408 DC motor rated at 1200 W, 2800 r/min, 38 N·m, and 48 V, which leaves a power margin for stable operation. Matching its rated speed to the 95.5 r/min wheel speed implies a reduction of about 29:1.

E. Rack-and-Pinion Steering

Steering uses a rack-and-pinion mechanism consisting of the steering motor, pinion, rack, tie rods, and ball joints (Table VI). The motor turns the pinion, which moves the rack laterally; the tie rods transmit this motion to the steering knuckles. All parts are made of 45 steel. The rack and pinion use standard involute teeth with a 20° pressure angle, and the pinion is quenched and tempered and ground for accuracy and low noise. Each tie rod ends in a ball joint with M10×1 threads and a lock nut, which lets the wheel steer freely without binding (Fig. 5).

Kinematics. To verify that the rack travel covers the ±30° steering range, we analyzed the linkage geometrically (Fig. 6). Each front wheel turns 30° from straight ahead, about 60° lock to lock. Applying the law of cosines to the steering-arm geometry ($O_1E \approx 81$ mm) gives a rack travel of 44.5 mm for a right turn and 44.1 mm for a left turn, 88.6 mm in total. The pinion–rack engagement length must exceed this, so 100 mm was chosen to leave a margin.

F. Steering Shaft Strength

The steering shaft carries the pinion and was checked at its critical section, where the pinion is mounted. Taking a shaft length $L = 150$ mm with the pinion 50 mm from the left bearing, radial loads of 1200 N (vertical) and 1000 N (horizontal)

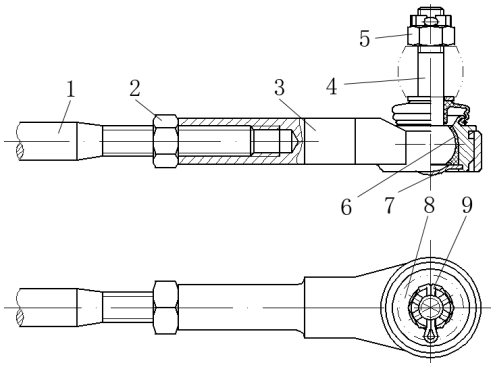


Figure 5. Tie-rod outer joint: 1 tie rod, 2 lock nut, 3 joint housing, 4 ball stud, 5 castle nut, 6 ball socket, 7 end cap, 8 steering arm, 9 cotter pin.

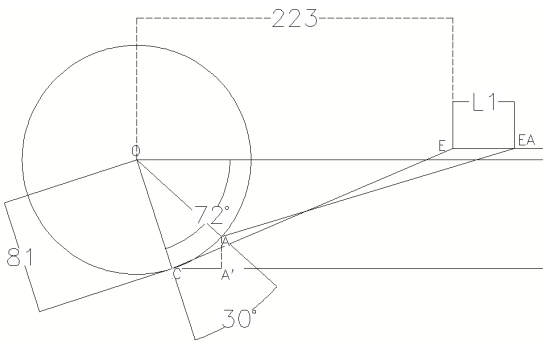


Figure 6. Kinematic sketch of the rack-and-pinion steering linkage for a 30° wheel turn (dimensions in mm).

give the support reactions in Table VII. The largest resultant bending moment is $M = \sqrt{M_x^2 + M_y^2} \approx 104{,}220$ N·mm, and the torque is $T \approx 14{,}432$ N·mm.

Combined stress. By the third strength theory,

$$\sigma_{ca} = \frac{\sqrt{M^2 + (\alpha T)^2}}{W}, \qquad W = \frac{\pi d^3}{32}, \tag{4}$$

with $\alpha = 0.3$. For $d = 25$ mm, $\sigma_{ca} \approx 67.9$ MPa, which exceeds the allowable 60 MPa for 45 steel. The diameter was therefore increased to $d = 30$ mm, giving $\sigma_{ca} \approx 39.2$ MPa < 60 MPa.

Fatigue. Treating bending and torsion as pulsating loads, the safety factors are

$$S_\sigma = \frac{\sigma_{-1}}{\frac{K_\sigma}{\varepsilon_\sigma \beta}\sigma_a + \psi_\sigma \sigma_m}, \quad S_\tau = \frac{\tau_{-1}}{\frac{K_\tau}{\varepsilon_\tau \beta}\tau_a + \psi_\tau \tau_m}, \tag{5}$$

$$S = \frac{S_\sigma S_\tau}{\sqrt{S_\sigma^2 + S_\tau^2}}, \tag{6}$$

with $\sigma_{-1} = 275$ MPa, $\tau_{-1} = 155$ MPa, $K_\sigma = 1.5$, $K_\tau = 1.4$, $\varepsilon_\sigma = 0.85$, $\varepsilon_\tau = 0.88$, $\beta = 1.0$, $\psi_\sigma = 0.1$, and $\psi_\tau = 0.05$ [13]. This yields $S \approx 7.58$, well above the allowable range of 1.3–1.5, so the shaft is safe for long-term operation.

VII. PICKUP MECHANISM

A. Requirements and Arm Type

The pickup mechanism must position and orient its end effector in space and rotate a full 360°. It must grasp rigid items

Table VII
STEERING SHAFT STRENGTH CHECK

Quantity	Value
Vertical reactions F_{y1}, F_{y2}	400 N, 800 N
Horizontal reactions F_{x1}, F_{x2}	333.3 N, 666.7 N
Max. resultant moment M	104,220 N·mm
Torque T	14,432 N·mm
σ_{ca} at $d = 25$ mm	67.9 MPa (fails, >60)
σ_{ca} at $d = 30$ mm	39.2 MPa (passes)
Bending / torsion safety S_σ, S_τ	7.6 / 72.3
Combined fatigue safety S	7.58 (>1.3–1.5)

such as bottles and cans and pick up light items such as bags and tissues, handling objects up to $20 \times 10 \times 15$ cm and at least 100 g. It also needs a locating device for accurate pickup and a light but strong structure.

Four common arm types were considered. A Cartesian arm is simple to control but cannot adjust its orientation. A cylindrical arm covers a wide workspace but has limited orientation adjustment. A spherical arm is more flexible but complex to build and maintain. An articulated arm made of revolute joints offers the highest dexterity. We selected a planar articulated arm, which combines a compact structure with flexible motion and suits small and medium payloads.

B. Structure

The arm consists of a base, a waist, an upper arm, a forearm, and a wrist, and carries the end effectors. The base is a hollow ZG200 cast-steel part, and the waist is a thin-walled ZG200-400 casting that keeps the arm stable during operation. The upper arm (800 mm) and forearm (600 mm) are made of high-strength aluminum alloy, and each is driven by a motor for pitching motion. The median in the arm design is 2400 mm wide, so the maximum reach was set to about 1200 mm, enough to cover the whole median from one side. The wrist has three degrees of freedom (rotation, pitch or yaw, and gripper roll), each driven by its own DC motor.

Two end effectors share the wrist through a changeover device:

- a motor-driven *gripper* for rigid items such as bottles, cans, and boxes;
- a *suction pump* for light items such as plastic bags, tissues, and film. It is a D15L-series micro diaphragm pump, in which an eccentric drive turns rotation into diaphragm reciprocation to create negative pressure.

The changeover device switches between them automatically or manually according to the litter type. A LiDAR helps locate litter and guides the arm to the target.

C. Transmission

Table VIII lists the transmissions. The wrist uses a pair of bevel gears (20 teeth, module 2 mm, face width 15 mm, backlash 0.15–0.2 mm) for pitch, and a 12 mm shaft keyed to the gripper for roll. The upper arm is driven directly through a 25 mm jaw coupling and key. The forearm uses the same coupling with a 1:5 transmission ratio for finer motion. The waist is driven through a 1:10 gearbox and a keyed 30 mm shaft, and the base is fixed with eight M16 bolts.

Table VIII
ARM TRANSMISSION COMPONENTS

Joint	Key parts and dimensions	Motor
Wrist	Bevel gears $z = 20$, $m = 2$ mm, width 15 mm; shaft ∅12×80 mm; keyway 6×6×20 mm	70BLD-48-300 BLDC + 30:1 planetary
Upper arm	Al alloy 50×30×5 mm section, 800 mm long; jaw coupling ∅25×50 mm; key 10×8×30 mm	FBL2360 BLDC, direct
Forearm	Al alloy 40×25×4 mm section, 600 mm long; jaw coupling ∅25×50 mm; ratio 1:5	FBL2360 BLDC
Waist	ZG200-400, OD 120 / ID 80 / H 150 mm; shaft ∅30×100 mm; key 12×8×40 mm; 8×M16 bolts	FBL2360 BLDC + 1:10 gearbox

Table IX
ARM MOTOR SELECTION

Joint	J_{tot} (kg·m^2)	Req. torque (N·m)	Motor (48 V)
Wrist	0.015	0.173 (inertial)	70BLD-48-300
Waist	16.91	18.4 (at motor)	FBL2360
Upper arm	–	13.05	FBL2360
Forearm	5.35	8.85	FBL2360

D. Motor Sizing

Each joint motor was sized from the inertia it drives and the required angular acceleration,

$$M = J_{\text{tot}}\,\varepsilon, \qquad M_{\text{motor}} = \frac{M}{i\,\eta}, \tag{7}$$

where i is the reduction ratio and $\eta = 0.95$ the transmission efficiency, with friction torque added where relevant.

Wrist. The design load is a $20 \times 10 \times 15$ cm, 100 g object on a 2 kg end effector, giving a wrist inertia of $J = 0.015$ kg·m^2. Reaching 330°/s (5.76 rad/s) in 0.5 s requires $\beta = 11.52$ rad/s^2 and an inertial torque of 0.173 N·m. A 48 V 70BLD-48-300 brushless motor (rated 1.5 N·m) with a 30:1 planetary gearbox was selected.

Waist. The waist carries the whole arm. Summing the forearm (4.22 kg·m^2), upper arm (6.53), arm motors (2.72), and waist itself (3.44) gives $J_{\text{tot}} = 16.91$ kg·m^2. With $\varepsilon = 7.854$ rad/s^2, the output torque is 132.81 N·m, or 18.4 N·m at the motor. A 48 V, 2880 W FBL2360 brushless motor was selected.

Upper arm and forearm. The same procedure gives required motor torques of 13.05 N·m for the upper arm and 8.85 N·m for the forearm (total inertia 5.35 kg·m^2). Both use FBL2360 motors, whose rated torque exceeds these values. Table IX summarizes the selections.

VIII. CONTROL PLATFORM

A. Design Principles

The control platform was designed around five principles:

- *Modularity.* Chassis control, arm control, and vision are decoupled so they can be developed, debugged, and upgraded independently.
- *Edge–cloud cooperation.* The cloud runs heavy AI inference while the robot handles real-time responses, including

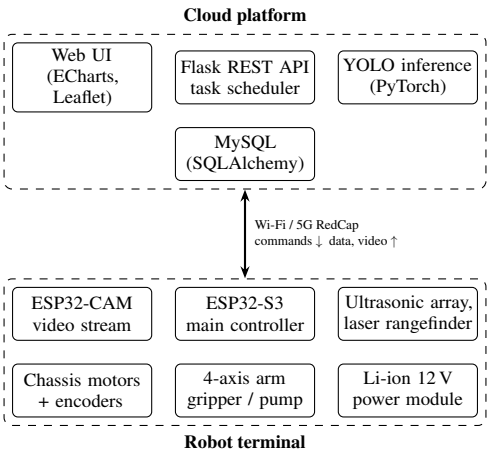


Figure 7. Edge–cloud architecture of the control platform.

chassis motion.

- *High availability.* The platform runs 24/7, handles faults automatically or manually, and the chassis diagnoses its own faults.
- *Ease of use.* Interfaces are simple, and the robot supports both manual remote control and autonomous cruising.
- *Extensibility.* The platform supports adding robots, hot-swapping AI models, and different hardware.

B. Architecture

Fig. 7 shows the edge–cloud architecture. The *cloud platform* is a web management system for task scheduling, AI inference, data storage, and visualization; it also issues motion commands to the chassis and arm. The *network layer* carries commands down and data up. The prototype uses Wi-Fi, and 5G NR RedCap is planned for low-power, low-latency field operation. The *robot terminal* uses ESP32 hardware with embedded firmware for sensing, chassis and arm control, data acquisition, and command execution.

The platform is organized in four layers. The *user layer* serves administrators, who manage the platform and view data, and field operators, who control robots manually and run detections. The *application layer* provides device management, task management, detection, data analysis, visualization, and manual control. The *service layer* hosts API, task-scheduling, AI-inference, data-storage, and chassis-control services. The *terminal layer* comprises the robot hardware, sensing, chassis control, arm execution, and communication modules.

C. Prototype Hardware

Table X lists the control prototype hardware. The ESP32-S3 runs chassis and arm motion control, data processing, and command parsing. The ESP32-CAM captures images and video, pre-processes them, and streams them to the cloud. The chassis is compact, at $400 \times 300 \times 150$ mm, with the arm mounted on top in the middle. The controller, motor drivers, and battery sit inside, and ultrasonic sensors are placed around the perimeter. Rubber anti-slip wheels with shock-absorbing pads protect the arm and electronics from road vibration.

Table X
CONTROL PROTOTYPE HARDWARE

Unit	Configuration
Main controller	ESP32-S3: chassis and arm control, command parsing
Vision	ESP32-CAM: image/video capture, pre-processing, streaming
Drive	2× 12 V/50 W DC gear motors, 1000-line encoders
Steering	12 V/30 W DC motor, gear transmission, ±30°
Wheels	∅120 mm rubber anti-slip, shock-absorbing pads
Sensing	Ultrasonic array (obstacles); laser rangefinder (grasp height)
Manipulator	4-axis planar articulated arm; gripper + D15L suction pump
Power	12 V/20 Ah Li-ion pack with level monitoring and alarm
Chassis size	$400 \times 300 \times 150$ mm (L×W×H)

Table XI
CODE ORGANIZATION OF THE CONTROL SYSTEM V1.0

Path	Content
`app.py`	Entry point and application factory
`config.py`	Global configuration
`api/`	Detection, robot, and statistics APIs
`inference/yolo_detector.py`	YOLO inference wrapper class
`database/models.py`	ORM models (users, tasks, robots)
`templates/`, `static/`	Front-end pages and assets
`Clean_Robot/src/`	ESP32-S3 and ESP32-CAM firmware
Python back end and algorithms	≈1,500–2,000 lines
Front end (HTML/JS/CSS)	≈1,000–1,500 lines
Total	≈3,500–5,000 lines

D. Software Stack

The back end is written in Python with Flask, a RESTful API, and SQLAlchemy ORM over MySQL. The front end uses HTML5 and JavaScript, with ECharts for charts and Leaflet for GIS maps. Detection uses PyTorch, Ultralytics YOLO, and OpenCV. The firmware is built with PlatformIO and the Arduino core for ESP32. The system runs on Windows 11 or Ubuntu 20.04, and the code base comprises roughly 3,500–5,000 lines. Detection is wrapped in a `YOLODetector` class that supports hot-swapping models and adjusting the confidence threshold without downtime.

Table XI shows how the code base is organized. The Flask application factory in `app.py` registers three groups of API blueprints, for detection, robots, and statistics. The detection API accepts images from the web form or from the REST interface and passes them to the detector. It saves annotated result images and stores each detection's label, confidence, and location in the database. The robot API handles device registration under a unique device ID, tracks online status through periodic heartbeats, records position, battery level, and working state (idle or working), and queues navigation targets and commands for each robot. The statistics API aggregates detections into category shares, time trends, and map hot spots for the front end. The firmware is a separate PlatformIO project for the two ESP32 boards. On the ESP32-S3 it implements PWM speed and direction control for the drive motors and reactive avoidance from the ultrasonic sensors; on the ESP32-CAM it implements real-time video streaming.

E. Interfaces

Table XII lists the core REST endpoints, which return JSON over HTTP. Upstream, the robot sends device status, position, video, detection results, battery level, arm state, rangefinder

Table XII
CORE REST API ENDPOINTS

Endpoint	Function
`GET /api/robot/list`	List robots (paging, filters)
`GET /api/robot/status`	Online state, battery, position
`POST /api/robot/control`	Forward, back, return home, etc.
`GET /api/task/list`	Tasks by status and time
`GET /api/task/detail`	Task details and detections
`POST /api/detect/upload`	Upload image, run detection
`GET /api/analysis/statistics`	Category, trend, spatial statistics
`POST /api/arm/control`	Target coordinates, effector, joints
`GET /api/arm/status`	Joint angles, effector mode
`POST /api/chassis/control`	Speed, steering angle, stop point
`GET /api/chassis/status`	Speed, steering, motor faults

readings, and chassis parameters. Downstream, the cloud sends tasks, remote-control commands, configuration, model updates, return-home commands, and arm and chassis commands. Arm commands and chassis emergency stops are prioritized so that critical instructions execute first. Between server and robot, a lightweight heartbeat and state-synchronization mechanism with automatic reconnection keeps devices tracked over weak networks.

F. Web Management System

The web front end has five main views.

- *Overview.* A GIS map of litter locations and robot positions and tracks, a ring chart of litter categories, arm statistics (grasp success rate, number of picks), and chassis statistics (speed, steering angle, mileage).
- *Robot management.* A device list with online state, battery, position, and arm and chassis status; remote control with click-to-set waypoints; and playback of historical tracks.
- *Task management.* A task list and drill-down details that compare the original and annotated images and show grasp coordinates, effector mode, speed, and stopping accuracy.
- *Statistics.* Category shares, 7- and 30-day trends, and heat maps of litter hot spots.
- *Manual operation.* Buttons and sliders for driving and steering, sliders for arm joints, and image upload for one-click detection, after which a grasp or stop-at-point command can be issued directly.

For field operations, the design further adds:

- *planning:* patrol plans by road section, frequency, time window, and no-go rules, with remote takeover, emergency stop, and geofencing;
- *event management:* filtering by time, section, category, confidence, and status; timeline playback; a choice of robot pickup, manual handling, or deferral for unreachable or dangerous cases; and same-spot re-inspection to confirm cleanup;
- *performance metrics:* response time, completion rate, recurrence rate, and litter per kilometer, shown on a situational dashboard with per-section KPIs;
- *a data loop:* human review of detections (correct, false positive, relabel, add missing), training data organized by scene (night, rain/fog, backlight, construction), and export in YOLO or COCO format for fine-tuning.

IX. PERCEPTION AND CONTROL ALGORITHMS

A. Litter Detection

Litter is detected with a one-stage YOLO detector [1]. A CNN backbone extracts features, and a feature pyramid network (FPN) [4] fuses them across scales, which matters here because targets range from 1 cm cigarette butts to 30 cm bottles. The final pipeline uses YOLO11 [7] in its nano variant (YOLO11n). During platform design, a fusion of YOLOv8 [5] and YOLOv10 [6] was also considered to balance accuracy and speed.

The training dataset covers 17 litter classes that span the survey's main categories: plastics (bottles, cups, bags, wrappers, Styrofoam), paper (paper bags, paper cups, cardboard, paperboard, Tetra Paks), glass and metal (glass bottles, colored glass bottles, cans), and other items (face masks, peel, rags, and piles of leaves). Images were enhanced and annotated with OpenCV. Each detection returns a category label, a confidence score, and a bounding box. For deployment, inference is accelerated with TensorRT [8] on the cloud server, and non-maximum suppression completes in under a second. Averaged over all classes, the model reaches 0.864 mAP@0.5 on the validation set, which meets the platform's 85% accuracy target; Section X reports the training procedure and per-class results. The platform also parses GPS coordinates from image metadata, geocodes them, and places each detection on the GIS map as a marker and in a heat map.

B. Hand–Eye Calibration and Coordinate Mapping

To grasp a detected item, its pixel position must be converted into arm coordinates. With the pinhole camera model [9], a point $\mathbf{P}_w = (X, Y, Z)$ in the world frame projects to pixel (u, v) as

$$s \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \mathbf{K} \, [\mathbf{R} \mid \mathbf{t}] \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}, \tag{8}$$

where $\mathbf{K}$ holds the camera intrinsics and $[\mathbf{R}|\mathbf{t}]$ the extrinsics. Through hand–eye calibration we derived the homogeneous transform ${}^{B}\mathbf{T}_{C}$ from the camera frame C to the arm base frame B, so a target located in the camera frame maps to the base frame as

$${}^{B}\mathbf{p} = {}^{B}\mathbf{T}_{C} \, {}^{C}\mathbf{p}. \tag{9}$$

The depth along the optical ray is fixed using the laser rangefinder, which also corrects the grasp height. The resulting Cartesian target is expressed in the arm's Denavit–Hartenberg (D–H) frames [10, 11]. The same mapping also gives the chassis a stopping point near the target.

C. Arm Inverse Kinematics

The arm has four joints: waist yaw θ_1, upper-arm pitch θ_2, forearm pitch θ_3, and wrist pitch θ_4. Given a target (x, y, z) in the base frame and a desired end-effector pitch ϕ, the waist angle is

$$\theta_1 = \operatorname{atan2}(y, x). \tag{10}$$

In the arm plane, let $r = \sqrt{x^2 + y^2}$. The wrist center is $r_w = r - L_4 \cos\phi$ and $z_w = z - z_0 - L_4 \sin\phi$, where z_0 is the shoulder height and L_4 the wrist-to-tool length. With upper-arm and forearm lengths L_2 and L_3 (800 and 600 mm in the full-scale design), the law of cosines gives

$$\cos\theta_3 = \frac{r_w^2 + z_w^2 - L_2^2 - L_3^2}{2L_2L_3}, \tag{11}$$

$$\theta_2 = \operatorname{atan2}(z_w, r_w) - \operatorname{atan2}(L_3 \sin\theta_3,\ L_2 + L_3\cos\theta_3), \tag{12}$$

$$\theta_4 = \phi - \theta_2 - \theta_3. \tag{13}$$

Forward kinematics checks the solution and monitors the end-effector position in real time. Joint limits are enforced during solving to protect the mechanism, and trajectories are smoothed by interpolation to avoid jerks. Joint speeds are slow and precise when grasping and faster when transporting.

D. Chassis Control

The chassis runs a closed loop of command reception, data acquisition, computation, and actuation with feedback. Speed and steering both use PID control [12]:

$$u(k) = K_p e(k) + K_i \sum_{j=0}^{k} e(j)\,\Delta t + K_d \frac{e(k) - e(k-1)}{\Delta t}, \tag{14}$$

where e is the error between the commanded and measured value (wheel speed from the encoders, or steering angle) and u is the PWM duty cycle sent to the motor driver. The robot cruises at 0.3–0.5 m/s and approaches a pickup point at no more than 0.1 m/s. Three steering modes are supported: front-wheel steering for normal cruising, rear-wheel steering for narrow areas, and synchronized steering for turning in place.

Path tracking combines the planned route on the GIS map, GPS position, and ultrasonic range data, adjusting speed and steering to follow the route and to detour around obstacles before rejoining it. For a pickup, the chassis decelerates, steers to the stopping point, and locks its drive and steering motors so the arm works on a stable base. A self-diagnosis routine monitors the motors, encoders, and ultrasonic sensors; if a motor stalls or a sensor fails, it stops the chassis, raises an alarm, and reports the fault to the cloud. Manual control always overrides autonomous control so that an operator can intervene quickly.

E. Coordinated Pickup Cycle

Fig. 8 shows the end-to-end pickup cycle:

1. While cruising, the ESP32-CAM captures and pre-processes images and uploads them over Wi-Fi.
2. The cloud runs detection and converts the pixel target into arm coordinates and a chassis stopping point.
3. The cloud sends the target, litter class, effector mode (gripper or pump), and stop command to the ESP32-S3 over the REST API.
4. The chassis stops near the target and locks.
5. The controller solves the inverse kinematics and corrects the grasp height with the rangefinder.

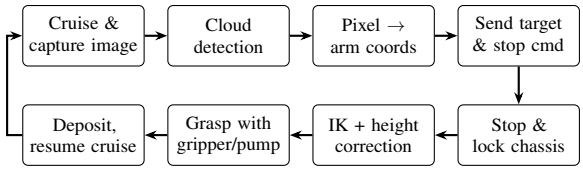


Figure 8. Coordinated "move–stop–pick–move" cycle between cloud, chassis, and arm.

6. The arm moves to the target, selects the effector by litter class, grasps the item, and reports the result.
7. The chassis unlocks, the arm deposits the item in the bin and returns to standby, cruising resumes, and the task status is updated in the cloud.

If the ultrasonic sensors detect an obstacle during a pickup, both chassis and arm stop immediately and resume once the obstacle has cleared.

X. Detection Model Training and Evaluation

This section reports how the litter detector was trained and how well it performs. All results come from the final training run, which used the Ultralytics framework [7]. Precision, recall, and mAP are computed on the validation split at an NMS IoU threshold of 0.7.

A. Dataset

The dataset contains 17 classes and roughly 20,000 annotated training instances (Fig. 9). It is strongly imbalanced. Plastic wrappers are the largest class at about 4,250 instances, followed by paper bags (about 2,950) and plastic bottles (about 2,100). Peel and rags are the smallest at about 250 each, a ratio of roughly 17:1 between the largest and smallest classes. The validation split holds 2,756 instances; Table XV gives the per-class counts.

The box statistics in Fig. 9 show two properties that matter for the robot. First, most objects are small: the width and height histograms peak below 0.1 of the image side, so at the 640-pixel training resolution a typical object spans only about 30–60 pixels. Second, box centers cluster near the middle of the image, which suggests many photographs were framed on the object. The robot camera will see litter anywhere in the frame and often near the edges, so this center bias is a domain gap to close with on-site data.

B. Training Setup

We fine-tuned YOLO11n, the smallest model of the YOLO11 family, starting from COCO-pretrained weights. YOLO11n has about 2.6 M parameters and 6.5 GFLOPs, small enough for a later move from the cloud to an embedded GPU on the robot. Table XIII lists the configuration. Training ran for 200 epochs with SGD and a linearly decaying learning rate, from 0.01 down to 1.5×10^{-4} at the last epoch, after a 3-epoch warm-up. Augmentation combined mosaic, random translation and scaling, horizontal flipping, HSV color jitter, RandAugment, and random erasing (Fig. 10). Mosaic was switched off for the final 10 epochs so the model could adapt to whole, unstitched images. Training was deterministic with

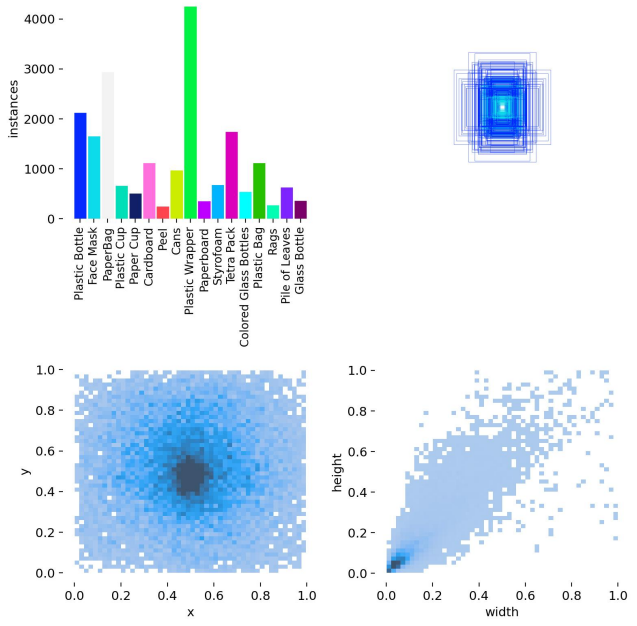


Figure 9. Training-set label statistics. Top left: instances per class. Top right: all boxes overlaid at a common center. Bottom: distribution of box centers and of box width versus height (normalized to image size).

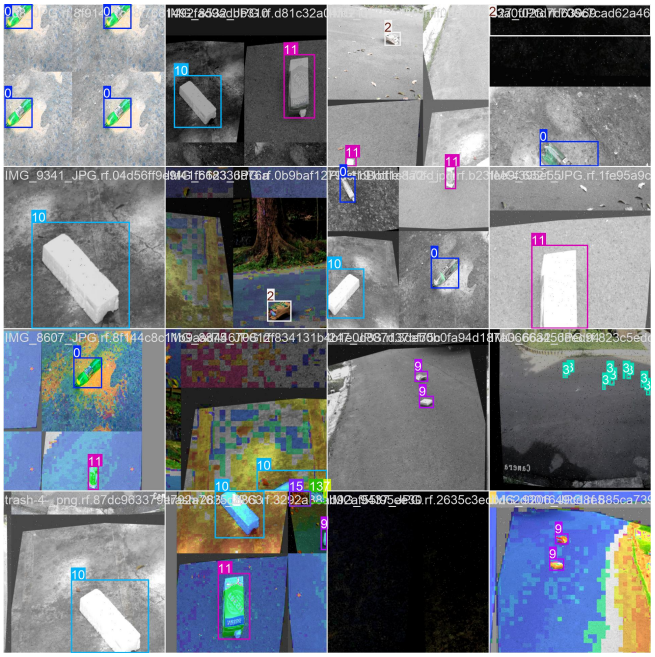


Figure 10. A training batch after augmentation. Mosaic stitches four images into one, and color jitter, RandAugment, and random geometric transforms vary appearance. Numbers are class indices.

a fixed seed and used automatic mixed precision. The run took 13,079 s (about 3.6 h, or 65 s per epoch); early stopping with a patience of 100 epochs was never triggered.

C. Convergence

Fig. 11 shows the losses and validation metrics over training. Table XIV gives mAP at selected epochs. Accuracy rises quickly at first and then saturates: mAP@0.5 reaches 0.743 by epoch 50 and 0.836 by epoch 100, which is 97% of its final value, and gains only 0.028 over the last 100 epochs. Training losses fall steadily to the end, while validation losses flatten after about epoch 150. The validation metrics never decline, so the model does not overfit within 200 epochs, but further epochs would bring little. Turning off mosaic at epoch 191 causes a visible step in the training classification loss (0.670 to 0.603) without changing the validation metrics. The best checkpoint, selected by the Ultralytics fitness score ($0.1\,\text{mAP@0.5} + 0.9\,\text{mAP@0.5:0.95}$), is the final epoch.

D. Overall Accuracy

At the final epoch the detector reaches a precision of 0.897, a recall of 0.799, an mAP@0.5 of 0.864, and an mAP@0.5:0.95 of 0.583. The mAP@0.5 result meets the platform's 85% target. The gap between the two mAP figures shows that boxes are reliable at a loose overlap criterion but less tight at strict ones. For grasping, this matters less than it might seem: the arm needs a good box center rather than exact edges, and the laser rangefinder corrects the grasp height (Section IX).

Fig. 12 shows how precision, recall, and F1 depend on the confidence threshold. Averaged over classes, F1 peaks at 0.84 at a confidence of 0.395, and recall reaches 0.93 when no threshold is applied. Raising the threshold to about 0.6 increases precision to roughly 0.95 while recall falls to roughly 0.70. This trade-off suggests two thresholds for field use. A low threshold (about 0.25) can mark possible litter on the GIS map for re-inspection, which costs nothing physically. A higher threshold (about 0.5–0.6) should gate the stop-and-pick cycle, because every false positive there costs a full stop, arm motion, and an empty grasp.

E. Per-Class Results

Table XV and Fig. 13 break the results down by class. Eight classes exceed 0.90 AP@0.5, led by paper bags (0.989), colored glass bottles (0.974), and Tetra Paks (0.970). These items have distinctive shapes, colors, or printed surfaces. Two classes fall far below the rest: piles of leaves (0.593) and transparent glass bottles (0.633). Plastic bags (0.798), plastic cups (0.838), Styrofoam (0.840), and peel (0.844) form a middle group.

Class size alone does not explain these differences. Rags and paperboard have only about 250–350 training instances each yet reach 0.929 and 0.852, while plastic bags, with about 1,100, reach only 0.798. Appearance and labeling matter more. A pile of leaves has no clear boundary, so its box is ambiguous, and it looks like the vegetation around it. Plastic bags are deformable and often thin and low in contrast. These are the same difficulties the field survey anticipated for litter in the median.

The two glass classes need a separate explanation. Clear, transparent bottles appear under the *colored glass bottles* label in the validation set (Fig. 14), so the two classes are not cleanly separated by color. The large gap between them (0.974 versus 0.633) may therefore reflect inconsistent annotation rather than a real difficulty with transparent glass. Merging the two into a

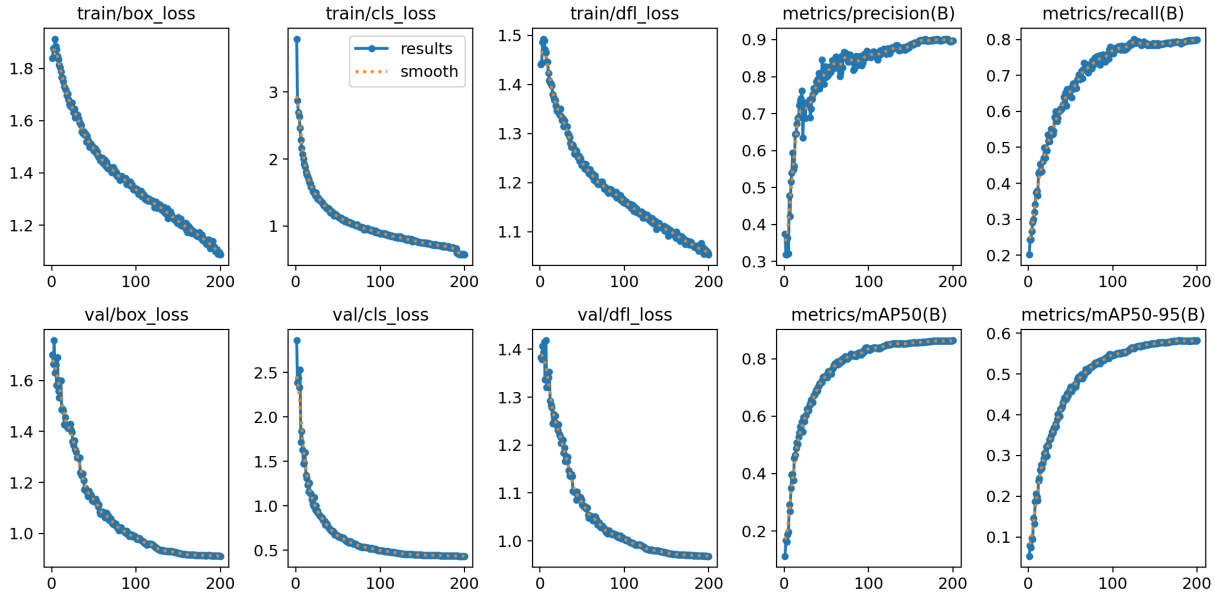
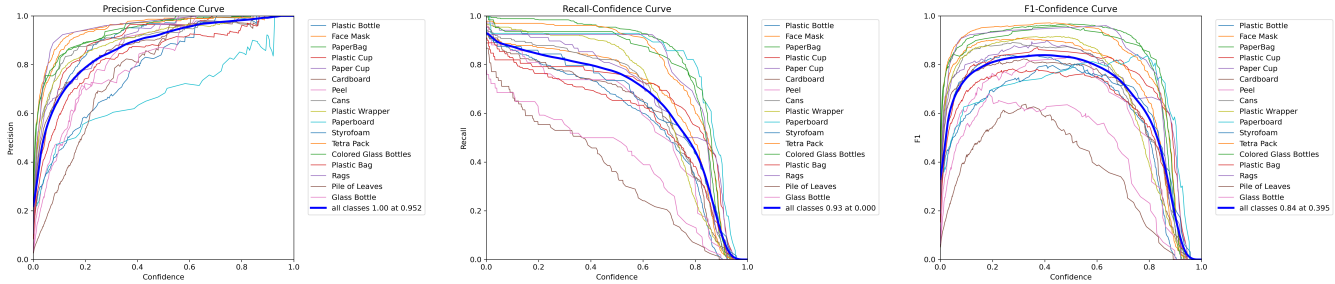


Figure 11. Training curves over 200 epochs. Top: training box, classification, and distribution focal (DFL) losses, and validation precision and recall. Bottom: validation losses, mAP@0.5, and mAP@0.5:0.95.



(a) Precision (b) Recall (c) F1

Figure 12. Precision, recall, and F1 against the confidence threshold for each class (thin lines) and averaged over classes (thick line). The averaged F1 peaks at 0.84 at a confidence of 0.395.

single glass class, or relabeling them consistently, should be tested before drawing conclusions about glass. For the robot, both classes go to the gripper, so a merged class loses nothing.

F. Qualitative Results

Fig. 14 compares ground-truth labels with the model's predictions on one validation batch. The detector finds every labeled object in these images, including small, distant cans and bottles, with confidence scores of 0.8–0.9, and its boxes closely match the annotations. These examples are mostly single objects on clean pavement and are easier than the median strip, so the quantitative results above remain the better measure of performance.

G. Error Analysis

The confusion matrix (Fig. 15) shows what kind of mistakes the detector makes. Of the 2,756 validation instances, 2,387 (86.6%) are detected with the correct class, 304 (11.0%) are missed, and only 65 (2.4%) are detected but assigned the wrong class. The detector also produces 283 false positives on background. The main weakness is therefore finding objects, not telling them apart.

The few class confusions are between visually similar materials. Thirteen cardboard items (7% of the class) are labeled paperboard, and 11 plastic wrappers are labeled paper bags. For the robot, such confusions are mostly harmless, because the class is used to choose the end effector. Cardboard and paperboard are both rigid enough for the gripper, and wrappers and paper bags are both light enough for the suction pump.

Misses are concentrated in the weak classes: 47% of leaf piles, 41% of glass bottles, and about 20% of plastic bags, peel, plastic cups, and plastic bottles are not detected. Background false positives are highest for plastic wrappers (58), plastic bags (30), Styrofoam (29), and cardboard (26). For Styrofoam this is 35% of all its predictions. Some of these may be real litter that was left unannotated in the validation images, so an annotation audit of these classes should come before further model changes.

Table XIII
DETECTOR TRAINING CONFIGURATION

Setting	Value
Model / initialization	YOLO11n / COCO-pretrained
Epochs / batch size	200 / 16
Input size	640×640
Optimizer	SGD, momentum 0.937
Learning rate	$0.01 \rightarrow 10^{-4}$ (linear)
Weight decay	5×10^{-4}
Warm-up	3 epochs
Loss weights (box/cls/DFL)	7.5 / 0.5 / 1.5
Mosaic	1.0, off for last 10 epochs
Geometric augmentation	translate 0.1, scale 0.5, h-flip 0.5
Color augmentation	HSV (0.015, 0.7, 0.4)
Other augmentation	RandAugment, random erasing 0.4
Precision / seed	AMP / 0 (deterministic)
Training time	3.6 h (13,079 s)

Table XIV
VALIDATION MAP DURING TRAINING

Epoch	mAP@0.5	mAP@0.5:0.95
1	0.113	0.053
10	0.399	0.198
25	0.602	0.340
50	0.743	0.468
100	0.836	0.548
150	0.855	0.575
200	**0.864**	**0.583**

H. Gap to the Field Scenario

These results measure the detector on its own dataset, not on the expressway median. Three gaps remain before the numbers can be trusted in the field.

- *Missing classes.* The dataset has no class for cigarette butts or tissues, which together make up 27% of median litter in the survey. Butts are also the smallest targets: a 1 cm butt seen from the robot's camera would cover only a few pixels at the 640-pixel input size.
- *Domain shift.* The images are not from Yantai Street. Most show litter on paved roads and sidewalks, photographed from above or by street cameras (Figs. 10 and 14), and the box statistics show centered, object-framed photographs. The robot will instead view grass, shrubs, and soil from a moving platform under glare, dusk, and rain.
- *No held-out test set.* The validation split was used both to select the checkpoint and to report results, so the figures are slightly optimistic. A separate test set, ideally collected on site, is needed for an unbiased estimate.

XI. SAFETY AND AUXILIARY FUNCTIONS

Working next to live traffic makes safety a first-class requirement. The design addresses it on four levels.

Warnings to road users. The operation warning system uses warning lights, reflective markings, and audible alarms so that drivers notice the robot. Lights are switched on for night work, and voice prompts ask pedestrians to give way near median openings. The shell is wear- and impact-resistant, with padded edges.

Table XV
PER-CLASS DETECTION RESULTS ON THE VALIDATION SET

Class	Train*	Val	AP@0.5	Correct	Missed
Paper bag	2950	433	0.989	0.98	0.01
Colored glass bottle	550	62	0.974	0.94	0.05
Tetra Pak	1750	230	0.970	0.97	0.03
Paper cup	500	68	0.935	0.88	0.07
Plastic wrapper	4250	455	0.935	0.92	0.05
Rags	250	26	0.929	0.92	0.08
Face mask	1650	255	0.908	0.87	0.12
Cans	950	173	0.905	0.89	0.09
Plastic bottle	2100	313	0.871	0.80	0.17
Cardboard	1100	199	0.866	0.79	0.12
Paperboard	350	28	0.852	0.93	0.04
Peel	250	42	0.844	0.76	0.21
Styrofoam	650	64	0.840	0.84	0.16
Plastic cup	650	72	0.838	0.79	0.18
Plastic bag	1100	199	0.798	0.75	0.21
Glass bottle	350	54	0.633	0.59	0.41
Pile of leaves	600	83	0.593	0.53	0.47
All classes	≈20,000	2756	**0.864**	0.87	0.11

*Approximate, read from the label histogram in Fig. 9. "Correct" and "Missed" are the fractions of validation instances detected with the right class and not detected at all, from the confusion matrix in Fig. 15.

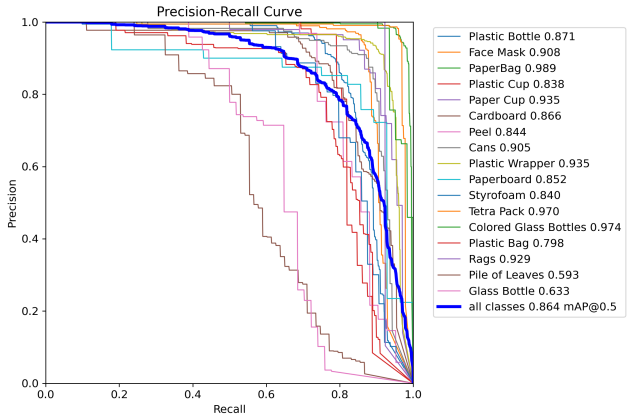


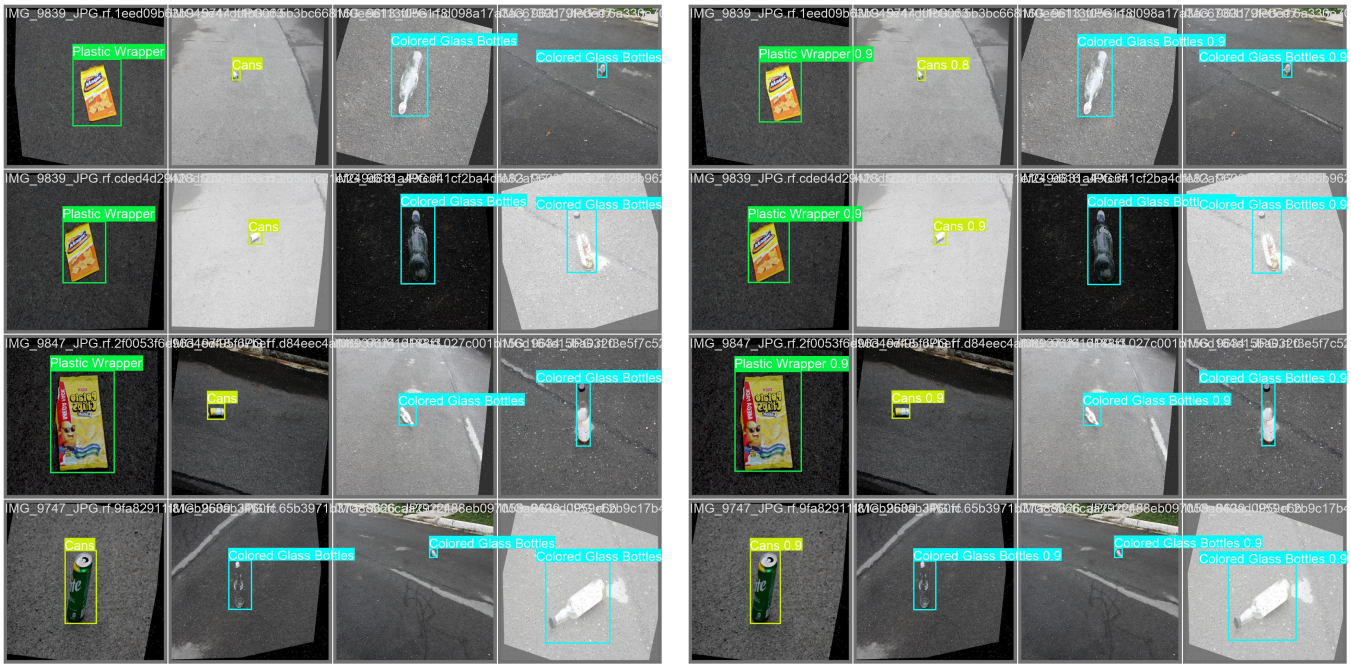
Figure 13. Precision–recall curves per class, with AP@0.5 in the legend. The thick line is the class average (mAP@0.5 = 0.864).

Motion safety. The robot keeps at least 0.5 m from the edge of the traffic lane and limits its speed to 4 km/h inside the green belt and 3 km/h in transition areas, as the survey recommends. Ultrasonic sensors around the chassis trigger an immediate stop of both chassis and arm when an obstacle appears, and emergency stops take priority over all other commands. Pedestrians and cyclists are always given right of way.

Fault handling. The firmware monitors the arm (motion anomalies, effector and rangefinder faults) and the chassis (motor stalls, steering faults, encoder and sensor failures). On any fault it stops, raises an alarm, and reports the details to the cloud. The battery module monitors charge and raises a low-battery alarm so the robot can return before it runs flat.

Human oversight. Operators can take over remotely at any time, stop the robot, send it home, or define no-go zones. Manual control always overrides autonomous control, and an unreachable or dangerous item can be deferred to manual handling rather than attempted by the robot.

Two auxiliary functions support continuous operation. A



(a) Ground-truth labels

(b) Predictions with confidence scores

Figure 14. Ground truth (a) and detector predictions (b) on a validation batch. Clear glass bottles in this batch are labeled as colored glass bottles, which illustrates the labeling inconsistency between the two glass classes.

Table XVI
PERFORMANCE TARGETS OF THE CONTROL PROTOTYPE

Metric	Target
Cloud command response	$\leq$1 s
Chassis command response	$\leq$0.5 s
Emergency-stop response	$\leq$0.2 s
Steering response / error	$\leq$0.3 s / $\leq 1^\circ$
Speed error	$\leq$0.1 m/s
Cruise / approach speed	0.3–0.5 / $\leq$0.1 m/s
Path-tracking error	$\leq$3 cm
Stopping accuracy	$\leq$5 cm
Detection accuracy	$\geq$85% mAP@0.5 (achieved: 86.4%)
Continuous operation	$\geq$8 h (terminal), 24/7 (platform)
Drive failure-free time	$\geq$100 h

bin-capacity reminder alerts operators when the storage bins are nearly full, so that collection can be scheduled. A *positioning system* combines GPS with encoder odometry, which places the robot on the GIS map and tags every detection with its location.

XII. Design Targets and Test Plan

Table XVI lists the performance targets set for the control prototype. Apart from detection accuracy, these are design targets that remain to be verified on hardware.

The test plan covers four areas:

- *Functional tests* of device and task management, detection, manual operation, visualization, arm inverse kinematics and effector switching, chassis speed, steering, tracking, and stopping, and coordinated operation.
- *Performance tests* of latency (commands, video, inference, arm and chassis response); detection accuracy under normal light, occlusion, and cluttered backgrounds; grasp accuracy and repeatability; long-duration stability; and concurrency with multiple robots and tasks.
- *Chassis tests* on asphalt, paving, and grass, measuring maximum climbing angle, battery endurance, and emergency-stop response.
- *Compatibility tests* across Windows and Ubuntu, Chrome, Edge, and Firefox, and different ESP32 boards, sensors, arms, and motors.

XIII. Project Execution and Outcomes

The project ran from May 2025 to May 2026 in three phases and finished on schedule:

- *Survey:* the two field surveys on Yantai Street;
- *Model training:* dataset collection, annotation, and cleaning, followed by several rounds of training and tuning, ending with the 200-epoch YOLO11n model reported in Section X;
- *Design:* chassis design (storage, locomotion, ground cleaning), arm selection, hardware selection, and design of the control and management software, including autonomous driving and data analysis.

The deliverables are:

- two survey reports, on the median environment and on litter types;
- three design documents, on the chassis and locomotion mechanism, the pickup mechanism, and the control platform;
- one software system, the *Intelligent Litter Detection and Cleaning Robot Control System V1.0* (completed Febru-

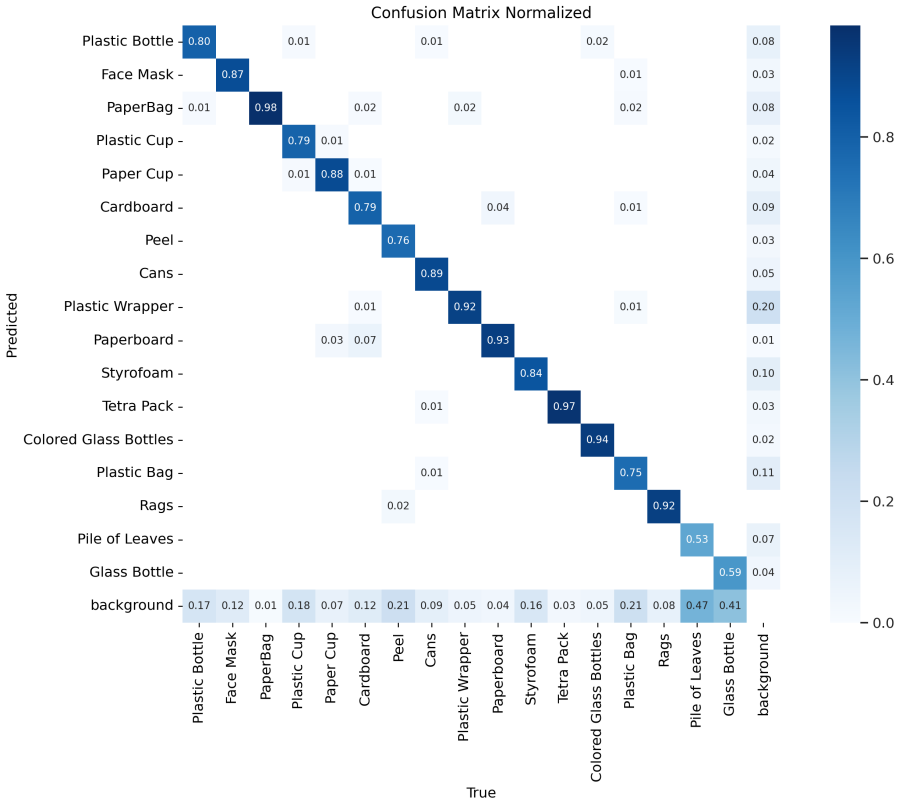


Figure 15. Normalized confusion matrix on the validation set (columns: true class; rows: predicted class). The bottom row gives the fraction of each class that was missed; the right column gives false positives on background.

ary 2026 and prepared for software copyright registration), which integrates litter recognition, robot operation and positioning, data analysis, autonomous driving with obstacle avoidance, and remote control;
- five individual reflection reports and a project summary report.

The work was carried out by a cross-disciplinary team. Students from mechanical engineering led the mechanism design. Hongyi Hao, from computer science, served as control-platform and robot-vision engineer, responsible for the detection system, hand–eye coordinate mapping, the Flask cloud platform, ESP32-S3 motion logic, and the communication link. Students from industrial and packaging design contributed to the appearance and structure.

XIV. Lessons Learned

Four lessons from the project are worth recording for future student teams.

- *Clear goals and phases.* The project kept one goal in view, an unmanned, fully automatic expressway litter robot. It divided the work into survey, model training, and design phases, each with defined tasks, deadlines, and owners. This structure is why every core task finished on time.
- *Survey first.* Surveying the real road before designing gave the detection dataset, mechanism dimensions, and software features a concrete basis and kept the design from drifting away from actual needs. Examples include the focus on small butts, the 1.2 m arm reach, and the time-window scheduling.
- *Focus on the hard problems.* Concentrating effort on the three key problems (avoidance and path following, recognition, and the arm) and on the interfaces between them kept the software and hardware designs consistent.
- *Cross-disciplinary teamwork.* Members from mechanical engineering, computer science, and design divided the work by expertise and resolved integration issues quickly. The team completed the project without dedicated funding, which showed a low-cost way to make progress.

XV. Limitations

The project leaves several gaps.

- *Survey coverage.* The surveys covered only Yantai Street, so other expressways with different layouts and traffic volumes are not represented.
- *Technical maturity.* Obstacle avoidance under sudden lane changes and extreme weather needs more work. The arm has not been optimized for tight spaces and unusual angles.
- *Detection.* The detector misses 11% of validation objects, and nearly half of leaf piles and transparent glass bottles. Its dataset has no cigarette-butt or tissue class, is not drawn from the expressway median, and has no held-out test set (Section X).

- *No physical prototype.* No dedicated funding or hardware was used, so no physical robot was built and no field trials were run. The feasibility and stability of the full system are therefore unverified, and cost control and maintenance planning for deployment are incomplete.
- *Design consistency.* The full-scale mechanical design and the reduced-scale control prototype use different power levels and drive layouts. They must be reconciled before a field machine is built.
- *Vision robustness.* Robustness under extreme lighting and cumulative localization error on uneven ground also need improvement.

XVI. Future Work

Next steps follow directly from these limitations:

- *Prototype and pilot.* Secure funding to build the physical robot, test it on a dedicated platform that simulates complex traffic and bad weather, and run a pilot on Yantai Street before extending to other expressways in Yangquan.
- *Perception.* Extend the dataset with multi-site and multi-condition samples, focusing on small and occluded litter. Add cigarette-butt and tissue classes from images taken on Yantai Street, and hold out an on-site test set. Audit annotations for the classes with many background false positives, and target the weakest classes with more data, higher input resolution or tiled inference for small objects, and class-balanced sampling. Add infrared thermal imaging for all-weather sensing. Move a lightweight detector onto the edge device to reduce bandwidth dependence; at about 2.6 M parameters, YOLO11n is a suitable starting point.
- *Navigation.* Fuse IMU data with SLAM for more accurate localization and path planning.
- *Manipulation.* Optimize the arm for confined spaces and difficult angles.
- *Deployment.* Develop cost-control, operation and maintenance plans, and seek collaboration with universities, research institutes, and companies.
- *Data and surveys.* Extend surveys to more sections, traffic levels, and regions, and set up a long-term mechanism that tracks changes in the roadside environment and keeps the data current.
- *Team.* Strengthen training in hardware integration and algorithm optimization, refine the division of roles, and recruit members who combine hardware and algorithm skills.

The planned path to deployment proceeds in stages. The first stage is a physical prototype tested on a simulation platform; the second, a pilot on Yantai Street with user feedback; and the third, wider deployment across Yangquan's expressways once the pilot is mature. Throughout, the hardware and software will be iterated based on field results and industry developments.

XVII. Conclusion

This project designed an AI-based robot for patrolling expressway medians and picking up litter, grounded in a field survey of a real road. The survey quantified the median environment and showed that plastics, paper, and cigarette butts dominate the litter, concentrated near intersections. The mechanical design specifies a rear-drive, front-steer chassis with a verified rack-and-pinion steering system and a planar articulated arm with a 1.2 m reach and dual end effectors. The control platform combines cloud-based YOLO11n detection, which reaches 0.864 mAP@0.5 over 17 litter classes, with real-time ESP32 control, hand–eye coordinate mapping, inverse kinematics, and PID chassis control, all managed through a GIS-enabled web system. Building and field-testing a physical prototype is the essential next step toward safe, unmanned expressway cleaning.

Acknowledgment

This work was supported by the Shanxi Provincial Undergraduate Innovation Training Program (Project No. S202514527040) and builds on the advisor's project "Research and Application of an Intelligent Litter-Picking Robot for Urban Expressway Median Strips." The team thanks Prof. Meiqing Zhao for her guidance.

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